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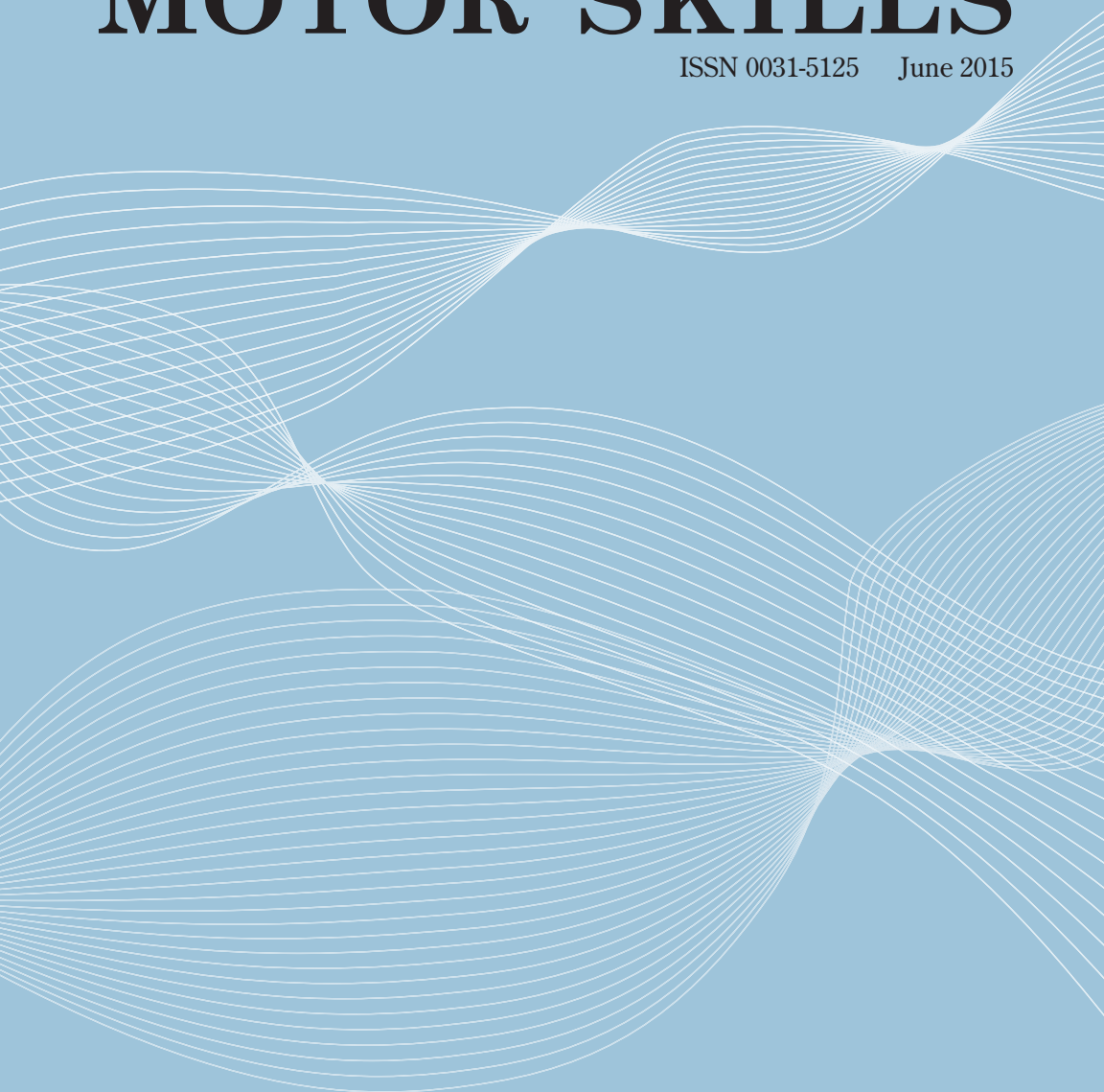
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PERCEPTUAL & MOTOR SKILLS

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THE EFFECTS OF SEX, TOPOLOGICAL STRUCTURE, AND TASK TYPE ON HYPERTEXT NAVIGATIONAL PERFORMANCE¹

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Summary.—Currently, almost all online materials use hyperlinks to provide users access to background, supplemental, or alternative information presented in context, greatly increasing the potential integration of information. However, a major problem is that people do not navigate hyperlinks effectively when the links become more topologically complex. Thus, identification of the variables that lead to navigational errors is necessary for the effective design of hyperlinks. Ninety-one participants (45 women, 46 men) were recruited for this experiment. All were college students and ranged in age from 19 to 23 yr. ($M=20.87$, $SD=1.02$). Navigational performance was examined in relation to sex, topological structure, and task type. A network topology with single-node task was superior to one with a linear topology under a single-node task condition, but equal to one with a linear topology under a multi-node task condition. Men navigated the linear topology with multi-node task and a network topology with a single-node task significantly faster than women, whereas no significant differences were observed under the other conditions. Sex interacted with topological structure and task type. This study extended the research in this domain by demonstrating an interactive effect among sex, topological structure, and task type on the navigational performance of users and can contribute to research regarding web page design.

A hyperlink allows users to access additional information, figures, or other material in many different ways, navigating to different informational items through links provided in a text (Nelson, 1974). Although hyperlinks extend users' options by offering the freedom to explore additional documents according to each user's needs, previous studies have indicated that, in practice, users do not benefit from the degree of freedom that hyperlinks provide (Conklin, 1987). Indeed, the major limiting factor of hyperlinks is that users often get lost or become disoriented due to the topological complexity of the links (Chang & Chen, 2011).

Two common topological structures in hyperlinks are linear and network (Shin, Schallert, & Savenye, 1994; Chen, Chang, & Chang, 2008). A linear topology is characterized by linear links, which guide the user with-

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in a more rigid structure, resulting in less disorientation. However, this kind of rigid structure results in potentially tedious back-and-forth and backtracking navigation. In contrast, a network structure provides users with all the functions and layouts on a single user interface, enabling them to visualize all of them simultaneously. Under these conditions, users can “jump” to different functions or units using hypertext features. However, this type of structure may result in cognitive overload as users attempt to find a specific single hypertext within the structure but do not know where to go or how to navigate the various types of information. Previous studies (McDonald & Stevenson, 1998; Lin, 2003) have attempted to identify the best hypertext topology by comparing the effectiveness of different topological interfaces for a specific task. These studies tested hypotheses based on the theory of spatial metaphor (Boechler, 2001; Lin, 2004), which assumes that the user's behavior while perusing hypertext is similar to their spatial processing in a physical environment. In general, the results regarding which topological structure is easier to use and maximally enhances navigation have been inconsistent. Ahmed, Coughlan, Edwards, and Morar (2009) argued that task type and individual difference are two major reasons for this inconsistency.

Task characteristics are one of the main contributors to differences in hypertext topology. A previous study indicated that successful design of specific task factors and their interaction with cognitive phenomena were key issues in designing hypertext that reduces disorientation (Chang & Chen, 2011). Another report (Chen, *et al.*, 2008) argued that a multi-node web hyperlink task, which requires cross-referencing information from more than one web page, is similar to the representation of network topology, hypothetically reducing search time and the number of pages opened. In contrast, a single-node task, which requires information only from a single web page, is similar to a linear topology and may therefore involve less memory load for the user to process information in single-node task, enabling faster navigation.

Hypothesis 1. Task type will significantly affect the impact of topological structure on web-browsing performance.

According to spatial metaphor theory, hypertext perusal behavior is similar to spatial processing in a physical environment (Kim, 1999). In addition to differences in individual wayfinding performance, significant sex differences have also been found, especially with respect to spatial capacity (Malinowski & Gillespie, 2001), which suggests a difference in the cognitive style of men and women that may result in performance differences in spatially related tasks. Previous studies regarding the navigation of the physical environment have suggested that men demonstrate a superior capacity to shift spatial strategies (Lawton & Morrin, 1999; Scali,

Brownlow, & Hicks, 2000) and visual working memory (Cornoldi & Vecchi, 2003; Coluccia & Louse, 2004).

Several studies have found that individual preferences for topics are significantly associated with spatial capacity (Hidi, 2001; Salmerón, Kintsch, & Cañas, 2006). Specifically, interest in a topic is associated with greater spontaneous attention and faster information processing (i.e., shorter search times; Hidi, 2000; McDaniel, Waddill, Finstad, & Bourg, 2000). A previous study (Chang, Huang, & Lin, 2011) tested the relationships between subjective preferences for seven topics (politics, international news, economics, cosmology, entertainment, vehicles, and tourism) and a web-browsing performance test. This study reported two main findings. First, they found a significant interaction between topic preference and sex; men preferred vehicles, but women disliked this topic, whereas women preferred cosmology, but men disliked it. Additionally, both men and women expressed an interest in tourism. Second, a significant interaction was found among web-browsing performance, topic preference, and sex; these were significantly associated with the male participants' web page-browsing performance but not with the female participants' performance. It is important to note that the effects of sex, task difficulty, and the topological structure of the web-browsing capabilities remain to be identified. Because the topic of tourism was of interest to both men and women, the use of this topic in the study may have disguised potentially significant interactions and effects due to differences in preferences for topics. However, previous studies (Lin, 2003; Bosco, Longoni, & Vecchi, 2004) did not include sex, topological structure, and task type, thereby precluding a realistic evaluation of navigational performance, which requires the inclusion of these factors and their interactions.

Hypothesis 2. Sex, task type, and topological structure will affect web-browsing performance.

METHOD

Participants

Ninety-one participants (45 women, 46 men) ranging in age from 19 to 23 years ($M=20.9$, $SD=1.0$) were recruited for the experiment. All were college students from the Department of Product Design of Ming Chuan University, Taiwan. They spent an average of 2.9 hr. ($SD=3.1$) engaged in daily Internet browsing. The participants were randomly assigned to one of the between-subjects (sex and topological structure) experimental conditions. A total of 45 participants (22 women, 23 men) were assigned to the linear topological interface condition, and 46 (23 women, 23 men) were assigned to the network topological interface condition. All participants were paid US \$5 per hour for their participation.

Materials and Design

Topic preferences.—As a previous study (Chang, *et al.*, 2011) had indicated that men favored and women disliked the topic of vehicles and that women favored and men disliked the topic of cosmology, tourism was chosen as the topic to avoid the potential interaction caused by topic preference.

Topological structures.—Each web page included between 400 and 447 Chinese Ming characters ($M=421$, $SD=14$) in 14-point type, yielding a total web content of 25,267 words. Most of the text was collected from the Internet and consisted primarily of Chinese characters; however, a few English words were used. The web pages were created using Hypertext Markup Language (HTML) based on two different hyperlink topologies: linear and network. Each of the two topological sets of web pages provided 20 hyperlink nodes.

The linear topological web page design featured each node in a fixed linear sequence. Each individual web page could be accessed by clicking on the “forward” and “backward” buttons at the bottom of the pages, which caused the next or previous page to be displayed. The links in the linear version were based on one parent node for every two child nodes, resulting in 20 nodes with 19 links (Fig. 1).

The network topological web-page design allowed each page to be accessed by clicking on its name in the text table at the bottom of the main web page. The pages were based on 20 nodes with 209 cross-referenced links, with each web page connected to every other page (Fig. 2).

Types of task.—The specific types of task were single-node, in which the task was to find a specific page related to each web topic that provided answers to the authors' questions, and multi-node, in which the task was

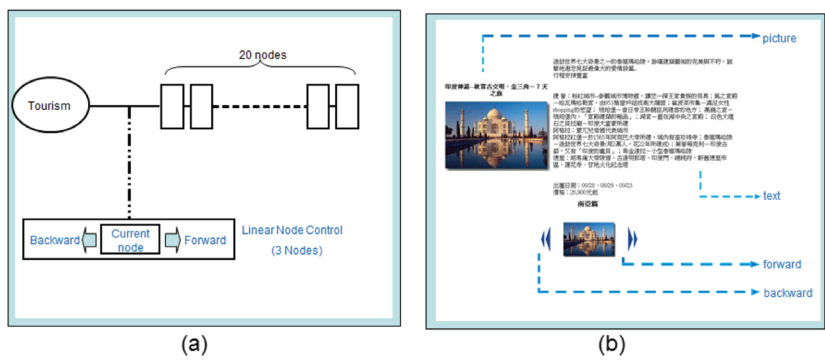


FIG. 1. Web design of the linear topological structure: (a) linear topological structure; (b) illustration of the interface and layout design.

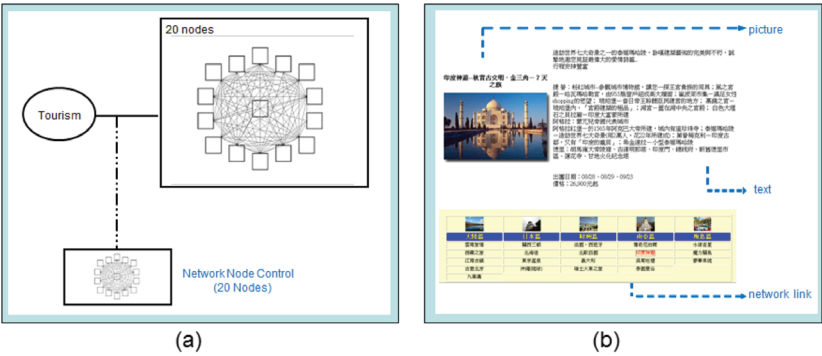


FIG. 2. Web design of the network topological structure: (a) network topological structure; (b) illustration of the interface and layout design.

to find the answers on three specific pages after browsing all the web pages for the assigned tasks.

Experimental Design

A mixed-factor $2 \times 2 \times 2$ multivariate analysis of variance (MANOVA) experimental design was employed in this study. The independent variables were sex, hyperlink topology (linear and network), and task type (single-node and multi-node). The dependent variables were the mean time for locating specific nodes. A within-subjects design was used for task type, and a between-subjects design was adopted for sex and topology. The experiment consisted of a total of six tasks: three tasks for each of the two types (single-node or multi-node). To avoid a learning effect and the use of different starting nodes for each task, the tasks were arranged using a Latin square, and each task started with the first page of the web site.

Procedure

All documents were run on Microsoft's Internet Explorer 9.0 browser and were displayed on a 17-in. LCD color monitor. The resolution was 1024×768 pixels, and the frame rate was 85 Hz. The activity of the participants was monitored throughout the experiment.

The participants were asked to learn to interact with the user interface by performing practice tasks. No time constraints were imposed during the practice session. The session ended when the participants were confident about their basic understanding of the interface hierarchy and the relationships among the various nodes.

After the participants were familiar with the web design, the experiment began. Each task was explained on a task card, and the participants

followed the order specified on the list to complete the navigational tasks. The participants were required to complete all the interaction tasks. The full experiment lasted approximately 1 hr. for each participant.

Data Analysis

Web-searching performance was measured by the time spent searching, starting when the task began and ending when the participants found the correct answer and provided their answer on the task cards. Each answer and task time were confirmed by the experimenter before the next task was presented. A mixed-design MANOVA ($2 \times 2 \times 2$) was performed to analyze the main effects and interactions of the variables. *Post hoc* comparisons were performed to identify the simple effects of the factors. The statistical analyses were performed using the Statistical Package for the Social Sciences (Version 20; IBM SPSS Statistic, Inc., Chicago, IL).

RESULTS

Overall Analysis of Navigational Performance

Table 1 presents the means and standard deviations for navigation time by sex, topological structure, and task type. The mixed-design repeated-model analysis indicated that the main effects of sex ($F_{1,87}=8.00$, $p<.01$, partial $\eta^2=0.08$), topological structure ($F_{1,87}=7.62$, $p<.01$, partial $\eta^2=0.08$), and task type ($F_{1,87}=227.54$, $p<.01$, partial $\eta^2=0.72$) were significant. Specifically, men spent less time than women completing the tasks, their navigation times were significantly faster using network topology than linear topology, and they performed single-node tasks more rapidly than multi-node tasks. The authors also found a significant two-way interaction involving task type $\times$ topology ($F_{1,87}=6.12$, $p<.05$, partial $\eta^2=0.07$) and a three-way interaction involving sex $\times$ task type $\times$ topology ($F_{1,87}=5.96$, $p<.05$, partial $\eta^2=0.06$).

Post hoc Analysis of Interactions

The *post hoc* test of the two-way interactions revealed that single-node tasks were performed faster than multi-node tasks regardless of whether

TABLE 1
DESCRIPTIVE STATISTICS FOR NAVIGATION TIME (SEC.) BY SEX, TOPOLOGICAL
STRUCTURE, AND TASK TYPE

Sex	Topological Structure	<i>n</i>	Task Type			
			Single-node		Multi-node	
			<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Women	Linear	22	23.4	5.47	42.1	10.20
	Network	23	17.6	5.04	36.4	10.11
Men	Linear	23	22.4	11.53	31.4	9.72
	Network	23	14.0	3.87	33.9	12.80

the topology was linear or network. Network topology was associated with faster task completion than linear topology under the single-node task condition, whereas no significant difference was found under the multi-node task condition.

Because a three-way interaction was also significant, *post hoc* tests were performed to identify the source of the effect among the experimental factors. Table 2 presents a summary of the *post hoc* comparisons of the three-way interaction for topological structure, task type, and sex, showing the difference between linear and network topology with regard to web-browsing performance according to sex and topological structure. As these pairwise comparisons indicate, there were five significant group levels identified (labeled a, b, c, d, e); “a” represents the minimum, while “e” represents the maximum search time. It was noted network topology consumed less time than linear topology under the single-node, but not the multi-node, condition. Table 3 shows the *post hoc* comparisons for the three-way interaction, with the same level system (five significant different groups, labeled a, b, c, d, e), indicating the difference between the web-browsing performance of men and women under topological structure and task conditions. As these pairwise comparisons show, men completed the tasks faster than women under two conditions, the linear topological condition with a multi-node task and the network topological condition with a single-node task, whereas no significant sex difference was found under the linear topology condition with a single-node task or the network topology condition with a multi-node task.

TABLE 2
THREE-WAY INTERACTION: *POST HOC* COMPARISONS BY BONFERRONI
TESTS ON TOPOLOGY, TASK TYPE, AND SEX

Treatment			n	M	M Difference, Subset for $\alpha = .05$				
Sex	Task Type	Topology			Group				
					a	b	c	d	e
Women	Single-node	Linear	22	23.4			◆		
		Network	22	17.6		◇			
	Multi-node	Linear	23	42.1					◆
		Network	23	36.6				◇	◇
Men	Single-node	Linear	23	22.4			◆		
		Network	23	14.0	◇				
	Multi-node	Linear	23	31.4				◆	
		Network	23	33.9				◇	

Note.—◆Linear and ◇Network topologies.

TABLE 3
THREE-WAY INTERACTION *POST HOC* COMPARISONS BY BONFERRONI
TESTS ON SEX, TOPOLOGY, AND TASK TYPE

Treatment			n	M	M Difference, Subset for $\alpha = .05$				
Topology	Task Type	Sex			Group				
					a	b	c	d	e
Linear	Single-node	Women	22	23.4			◆		
		Men	23	22.4			◇		
	Multi-node	Women	22	42.1					◆
		Men	23	31.4				◇	
Network	Single-node	Women	23	17.6		◆			
		Men	23	14.0	◇				
	Multi-node	Women	23	36.6				◆	◆
		Men	23	33.9				◇	◇

Note.—◆Women and ◇Men.

DISCUSSION

Interaction Between Topological Structure and Task Type

As noted in the Introduction, this study assessed the effects of different topologies on web-page browsing performance. The main effects for sex, topological structure, and task type were significant, as was the interaction between topological structure and task type. This finding is consistent with the argument that task type has a significant effect on navigational performance (Bosco, *et al.*, 2004; Darejeh & Singh, 2013). The significant interaction indicates that it is essential to specify the characteristics of the task when testing the effect of topological structure. The present study extended the findings of previous studies regarding hypertext navigational performance (Conklin, 1987; Ainley, Hidi, & Berndorff, 2002; Salmerón, *et al.*, 2006) by demonstrating that different topological structures and task types influence the navigational performance of users. Hypothesis 1 was supported by the significant interaction between task type and topological structure on web-browsing performance.

Interactive Effects of Sex and Other Factors

Figure 3 shows the three-way interaction of sex with topological structure and task type. Two findings should be noted. First, as mentioned in the discussion of the *post hoc* comparisons of the three-way interaction analysis among topological structure, task type, and sex, navigation performance was significantly faster with the network than with the linear topological structure under only the single-node task condition, whereas no significant difference was observed under the multi-node task condition. This differs

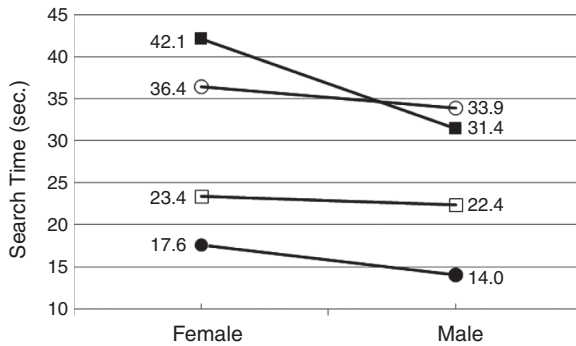


FIG. 3. Effect of sex on web search time by topology and task type (□ Linear/single node; ■ Linear/multi-node; ● Network/single node; ○ Network/multi-node).

from a previous study, in which linear topology was associated with better performance than network topology under a single-node task condition and that network topology may be associated with better performance than linear topology under a multi-node task condition (Lin, 2004; Chang & Chen, 2011). One possible reason for this discrepancy is that the previous studies did not include the effect of sex in its experimental design.

Second, it is important that men performed significantly better than women on the multi-node tasks with linear topology and on the single-node task with network topology, whereas no significant differences were observed under the other conditions (i.e., the single-node task with linear topology and the multi-node task with network topology). One possible reason for the effect of sex on performance with network topology and a single node is the increased number of linking nodes in the multi-node compared with the single-node task, which required more working memory. Thus, the effect of sex appeared when linear topology was used with a multi-node task. Because fewer linking nodes were needed for a linear topology with a single-node task and for a network topology with a multi-node task than for the other two conditions, less working memory was needed under the former conditions. This result is similar to those of a previous study when the task difficulty reached the limits of subjects' wayfinding capacity (Loring-Meier & Halpern, 1999; Graff, 2005; Coluccia, Bosco, & Brandimonte, 2007; Chen, Chang, & Chang, 2009). The present study found that the main effect of sex significantly interacted with topological structure and task type. Hypothesis 2 was also supported by the interaction among sex, task type, and topological structure.

Conclusion

Based on the foregoing, it was concluded that topological structure and task type interacted with sex. The design of new hypertext interfaces

should involve not only planning related to topological structure but also consideration of environmental and individual differences. In terms of the design of hypertext topology, comparison of the hypertext topological structure in the absence of consideration of task type or other related user-centered factors can be problematic. As the present study demonstrated, network topology is superior to linear topology, but factors related to sex and task type should be considered when applying this conclusion. This study extended previous findings by demonstrating the interactive effect of sex, topological structure, and task type on the navigational performance of users. The study results related to sex, topological structure, and task difficulty can benefit research regarding web page design.

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